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ORIGINAL ARTICLE

Role of structural relaxation and non-lattice oxygen in solution-processed ultra-thin-body crystalline indium oxide transistors

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Abstract
The introduction of ultra-thin-body (UTB) oxide semiconductors offers an attractive route toward back-end-of-line compatible electronics and advanced display backplanes because of their immunity to short-channel effects, low leakage current, and high process compatibility. However, compared with conventional thick oxide channels, UTB devices often exhibit degraded mobility and stability, and the trade-offs between key performance indicators become more pronounced. Here, we demonstrate the fundamental roles of structural relaxation and non-lattice oxygen (O_{NL}) in governing the performance–stability trade-off of sol–gel UTB indium oxide (In_2O_3) thin-film transistors (TFTs) through modulation of the annealing time (t_A). As t_A increases, structural relaxation progresses, as evidenced by contraction of the d-spacing and an increase in crystallite size. In contrast, the O_{NL} component shows non-monotonic evolution, with a decrease at intermediate t_A followed by a rebound at longer t_A. The t_A dependences of subthreshold swing (SS) and mobility correlate with the evolution of O_{NL}, which highlights its significant impact on device characteristics. Under strong positive bias stress, increasing t_A continuously and effectively suppresses overall device degradation. Notably, this stability trend differs from that of the initial electrical characteristics, which supports that the interplay between t_A-driven structural relaxation and O_{NL} represents a primary origin of the performance–stability trade-off in UTB TFTs. As a result, an optimal t_A processing window is identified to balance electrical performance and stability. Annealing-time engineering of UTB oxide TFTs, therefore, provides both physical insight into degradation pathways and parameter trade-offs, and a practical process optimization strategy for applications constrained by thermal budget.

Keywords Ultra-thin body · Structural relaxation · Non-lattice oxygen · Performance–stability trade-off · Annealing time

1 Introduction

Oxide thin-film transistors (TFTs) are core switching and driving elements in active-matrix displays and have also emerged as leading candidates for back-end-of-line (BEOL)-compatible electronics [1, 2]. This positioning stems from their high process compatibility with large-area and CMOS fabrication routes, intrinsically low off-state leakage enabled by wide bandgap semiconductors, and the feasibility of low thermal-budget processing, which is required to avoid thermal damage to pre-fabricated underlying layers in BEOL and monolithic three-dimensional (M3D) integration [3]. Within oxide semiconductors, indium oxide (In_2O_3) is particularly attractive because its s-orbital–dominated conduction band supports efficient electron transport, which enables high current drivability and fast switching behavior

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desirable for advanced backplanes and integrated electronics [4].

Despite these advantages, In_2O_3 TFTs often face practical limitations associated with oxygen-related instability. In_2O_3 exhibits a relatively low oxygen vacancy (V_O) formation energy compared with many other oxides, which can promote V_O formation and lead to excessive electron concentration [5, 6]. Although such high carrier density benefits current conduction, it can simultaneously deteriorate switching characteristics and gate controllability, yielding undesirable negative shifts in threshold voltage (V_{TH}), increased device-to-device variability, and high subthreshold swing (SS) [7]. Moreover, oxygen-related defects can evolve under electrical stress, which makes it difficult to achieve both high electrical performance and robust bias stability required for industrial circuit operation. This challenge arises because electrical characteristics and bias stability often exhibit a trade-off relationship and directly affect the performance and lifetime of electronic devices and integrated systems [8].

In this context, channel thickness scaling in oxide semiconductors is becoming increasingly important. An ultra-thin-body (UTB) channel provides a direct route to short-channel-effect (SCE) immunity because reducing the semiconductor body thickness enhances the gate controllability to modulate the channel potential in the vertical direction [9]. This behavior is explained by the concept of natural length, which indicates that shorter characteristic lengths reduce sensitivity to SCEs [10]. In other words, UTB scaling can improve switching characteristics and V_{TH} controllability of high-mobility channel materials while simultaneously mitigating SCEs that emerge as device dimensions shrink. Such scaling advantages are technologically significant for high-density integration and advanced backplanes, where channel length (L) must be reduced without sacrificing controllability and uniformity [11].

At the same time, UTB operation is accompanied by a well-known trade-off between thickness and performance or stability. As the channel becomes extremely thin, interface-related scattering becomes more influential, and the role of subgap states in channel formation becomes more pronounced because a larger fraction of the conduction path is confined near the interface region [12]. These effects often lead to degraded SS, reduced mobility, and weakened bias reliability or environmental stability. Therefore, UTB In_2O_3 TFTs require deeper insight into the factors that limit electrical performance and reliability, together with carefully designed process strategies that can control these limiting factors without introducing additional penalties.

Thermal annealing is one of the most effective and widely adopted approaches to tune oxide semiconductors [13, 14]. For both solution-processed and vacuum-deposited oxides, annealing is critical to densify the metal–oxygen network, reduce structural defects, and promote crystallization. These structural evolutions can strongly influence charge transport through changes in grain boundary density and associated potential barriers, and they can also modify oxygen-related bonding configurations that are frequently linked to subgap states and charge trapping in oxide semiconductors [15, 16].

In UTB channels, this coupling becomes more critical because subtle changes in structural disorder or oxygen bonding can affect a substantial fraction of the channel volume and manifest directly in TFT characteristics. Many studies have primarily addressed the effects of annealing temperature on semiconductor properties [17, 18]. However, next-generation applications, such as plastic-substrate-based organic light-emitting diode displays and M3D integration, face strict limitations on annealing temperature [19–21]. By contrast, fundamental understanding of semiconductor properties, particularly in UTB systems, as a function of annealing time (t_A), remains limited. Therefore, beyond observing performance changes with semiconductor t_A, it is essential to establish how t_A regulates UTB semiconductor properties at a fundamental level. Based on this understanding, identifying device trade-offs and proposing optimal processing conditions becomes possible.

In this study, we investigate fundamental changes in the semiconductor film as a function of t_A in crystalline UTB TFTs with a 4.5 nm In_2O_3 channel thickness. The In_2O_3 films were thermally crystallized at 370 °C, and t_A was varied from 5 to 180 min. Structural and chemical trends in the UTB In_2O_3 films were characterized as a function of t_A, and their correlations with TFT electrical properties were analyzed. Bias stability was further evaluated under negative bias stress (NBS) and positive bias stress (PBS) conditions to clarify how t_A modulates bias-induced performance changes in UTB In_2O_3 TFTs, in relation to the structural and chemical properties of the films. Based on the evolution of densification and rebound behavior of non-lattice oxygen (O_{NL}) with t_A, this study identifies an optimal t_A range that balances electrical performance and bias stability, which provides practical guidance for optimizing UTB In_2O_3 transistors.

2 Materials and methods

2.1 Preparation of oxide precursor solutions

In_2O_3 and yttrium oxide (Y_2O_3) precursor solutions were prepared to serve as the semiconductor and capping layers of the TFTs. For the In_2O_3 precursor, indium nitrate hydrate ($In(NO_3)_3 \cdot xH_2O$, Sigma-Aldrich) was dissolved in 2-methoxyethanol (2-ME) to obtain a 0.10 M

solution. For the Y_2O_3 precursor, yttrium nitrate tetrahydrate ($Y(NO_3)_3 \cdot 4H_2O$, Sigma-Aldrich) was dissolved in 2-ME to obtain a 0.05 M solution. Each solution was stirred at 35 °C for 24 h, which yielded optically transparent and homogeneous precursor solutions.

2.2 Fabrication of devices

Y_2O_3-capped In_2O_3 TFTs with a bottom-gate/top-contact configuration were fabricated [19]. The ultrathin Y_2O_3 capping layer was introduced to suppress oxygen and moisture adsorption and desorption at the oxide surface and to mitigate excessive electron generation in the In_2O_3 channel. P-type Si wafers with a 100 nm-thick dry thermally grown SiO_2 layer were used as the gate electrode and gate dielectric, respectively. The substrates were cleaned through sequential ultrasonication in acetone, isopropyl alcohol, and deionized water for 10 min per step, followed by ultraviolet–ozone treatment to remove organic residues and improve surface wettability. The In_2O_3 layer was deposited via spin coating, and isolated channel patterns were defined using a photopatterning process reported previously [22].

After pattern formation, the In_2O_3 thin film was pre-annealed at 100 °C for 5 min, followed by post-annealed at 330 °C for various t_A ranging from 5 to 180 min. Subsequently, the Y_2O_3 capping layer was annealed at 330 °C for 30 min. All annealing processes were performed using a hot plate under atmospheric environment. To prevent the influence of the ramping rate, samples were placed or removed while the hot plate was fixed at the target temperature. Finally, 50 nm-thick Al source and drain electrodes were deposited through thermal evaporation. The channel length (L) and channel width (W) were 100 μm and 1000 μm, respectively.

2.3 Analysis of UTB In_2O_3 films and TFTs

The structural properties of UTB In_2O_3 films were examined using transmission electron microscopy (TEM). The grazing-incidence X-ray diffraction (GIXRD) measurements of UTB In_2O_3 films were performed with a fixed incident angle of $\omega = 0.5°$ over a 2θ range of 10–80° using Cu Kα radiation. The In_2O_3 (222) diffraction peak was fitted using a pseudo-Voigt function after linear background subtraction, and the fitted peak center was used to calculate the d-spacing based on Bragg's law. To quantify the uncertainty in the small d-spacing variation, the fitting was repeated using slightly different fitting windows around the (222) peak, and the standard deviation of the extracted d-spacing values was used as the error bar. XRR measurements were performed to investigate the film density of UTB In_2O_3 films. Oxygen bonding environments in UTB In_2O_3 were analyzed using X-ray photoelectron spectroscopy (XPS) with an Al Kα source ($h\nu = 1486.6$ eV). Electrical characteristics were measured using a semiconductor parameter analyzer (Keithley 4200 A-SCS).

All electrical measurements, including bias-stress tests, were conducted in the dark at room temperature and ambient pressure. Transfer characteristics were recorded by sweeping the gate voltage (V_G) from −25 V to 25 V with a step of 0.2 V, while fixing the drain voltage (V_D) at 30 V to probe the saturation regime. NBS and PBS were performed by applying stressed gate voltages (V_{STR}) of −40 V and +40 V, respectively, for 1000 s, with V_D fixed at 0 V.

V_{TH} was determined using the constant-current method and defined as the V_G at which the drain current (I_D) equaled 10 nA × W/L. The subthreshold swing (SS) was extracted from the reciprocal slope of the $\log_{10}(I_D)$-V_G curve over the I_D range from 10^{-9} to 10^{-8} A. The saturation mobility (μ_{SAT}) was extracted at $V_G = V_{TH} + 10$ V and $V_D = 30$ V using:

$$\mu_{SAT} = \frac{2L}{WC_{ox}} \cdot \left(\frac{\partial \sqrt{I_D}}{\partial V_G} \right)^2 \quad (1)$$

Here, C_{ox} denotes the gate capacitance per unit area.

3 Results and discussion

3.1 Annealing-time-modulated structural relaxation and O_{NL} rebound in UTB In_2O_3

The bottom-gate, top-contact TFT architecture with In_2O_3 and a Y_2O_3 capping layer is illustrated in Fig. 1a. A representative cross-sectional TEM image of the In_2O_3/Y_2O_3 stack with an t_A of 30 min confirms that the intended UTB structure is successfully realized using the chemical solution deposition approach. The thickness of the UTB In_2O_3 and Y_2O_3 layers is approximately 4.5 and 2.5 nm, respectively. The fast Fourier transform (FFT) patterns provide evidence of crystalline ordering in both UTB In_2O_3 and Y_2O_3, which supports that thermal annealing yields crystalline In_2O_3 and Y_2O_3 networks even at nanometer-scale thickness. This structural analysis confirms that the UTB semiconductors used in this study are not fully amorphous but instead consist of a highly crystalline oxide network.

The structural characteristics of UTB In_2O_3 as a function of t_A were quantified using XRD analysis. The diffraction peak at approximately $2\theta = 30.7°$ corresponds to a representative reflection of crystalline In_2O_3. Variations in the peak position and shape reflect structural relaxation and grain growth. Two parameters, the interplanar distance (d-spacing) and the crystallite size (D), were extracted from the

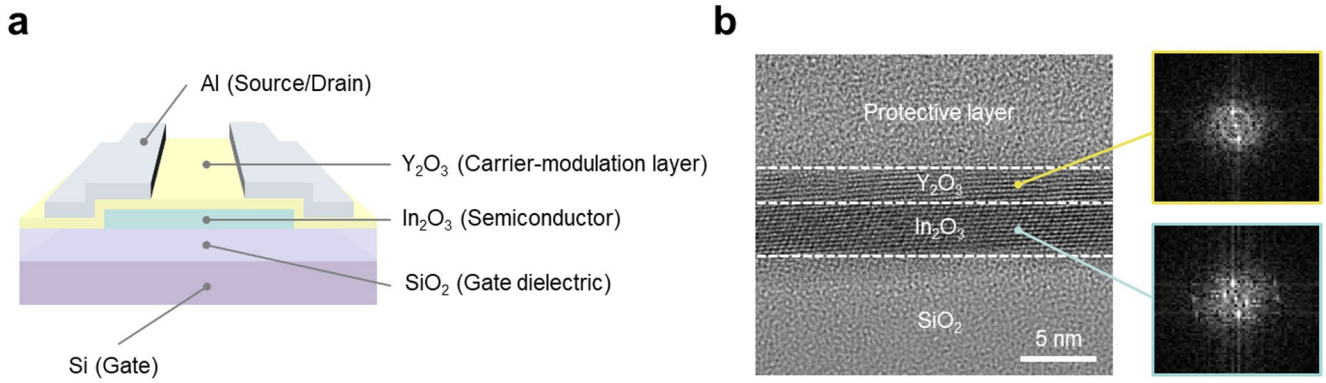


Fig. 1 (**a**) Schematic illustration of the bottom-gate, top-contact UTB In_2O_3 TFTs with a Y_2O_3 capping layer. (**b**) Cross-sectional TEM image (left) and corresponding FFT patterns (right), confirming uniform formation of the crystalline UTB In_2O_3/Y_2O_3 stack

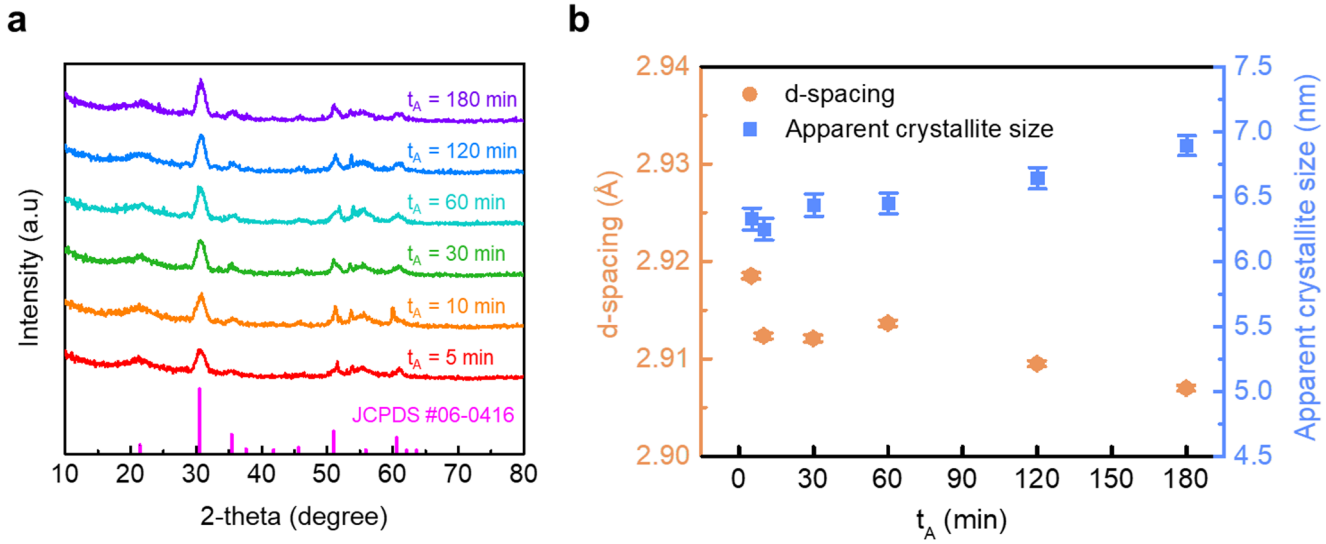


Fig. 2 (**a**) XRD patterns of UTB In_2O_3 annealed at different t_A. (**b**) Extracted d-spacing and Apparent crystallite size D from the XRD patterns of UTB In_2O_3 annealed as a function of t_A

XRD spectra. The d-spacing was calculated using Bragg's law [23]:

$$n\lambda = 2d\sin\theta \tag{2}$$

Here, λ and θ denote the X-ray wavelength and the incident angle relative to the X-ray crystal plane, respectively. The crystallite size was estimated using the Scherrer relation [24]:

$$D = K\lambda/(\beta\cos\theta) \tag{3}$$

Here, K and β represent the shape factor and the full width at half-maximum of the diffraction peak in radians, respectively. The crystallite size estimated from the Scherrer relation is presented as an apparent coherent diffraction domain size and is used mainly to compare relative changes with t_A. Notably, since the extracted size is on the order of several nanometers and is comparable to the UTB thickness and instrumental and strain broadening were not independently removed, it is advisable to view this as apparent coherent domain sizes for relative comparison rather than as absolute grain sizes.

The (222) d-spacing of cubic In_2O_3 is typically reported to be approximately 2.92 Å, and a comparable value was obtained in this study, as shown in Fig. 2b [25]. The extracted d-spacing decreases gradually with increasing t_A. Although the absolute variation is small, its significance is supported by atomic-scale sensitivity and the presence of a clear monotonic trend. The t_A-dependent d-spacing indicates that continuous structural relaxation and densification occur with increasing t_A. X-ray reflectivity (XRR) was additionally performed to quantify t_A-dependent densification of the In_2O_3 films. The extracted film density increases with increasing t_A, indicating progressive condensation and densification (Figure S2). These behaviors are attributed to

thermal removal or rearrangement of loose or incomplete metal-oxygen-metal (M-O-M) bonds, residual ligands, organics, and free volume remaining from the initial sol-gel process [26]. The reduction in d-spacing suggests that precise control of t_A promotes the formation of densely interconnected M-O-M networks, which enhances the structural robustness of the semiconductor films against external stimuli or stress. The crystallite size D also increases with increasing t_A, which indicates continued grain growth. In oxide semiconductors, a larger grain size generally reduces grain boundary density and lowers grain-boundary-related potential barriers, which can enhance electron transport and operational stability.

The influence of t_A on atomic bonding states was further investigated using XPS. Notably, the O 1s XPS spectra were acquired from In_2O_3 single films to exclude any oxygen contribution from the Y_2O_3 capping layer. Figure 3a presents the O 1s spectra of crystalline UTB In_2O_3 films as a function of t_A. Although deconvolution of the O 1s peak is widely used to track changes in oxygen bonding states in oxide materials, recent studies have emphasized that assigning individual deconvoluted peaks to specific chemical species without complementary evidence can be controversial [27–29]. In particular, direct attribution of the 531–532 eV component solely to V_O remains debated. For this reason, the O 1s spectra in this study were decomposed into two components: lattice oxygen (O_L) and non-lattice oxygen (O_{NL}). The spectral contribution near 533–534 eV, commonly associated with molecular H_2O adsorption, was negligible, and scenarios dominated by physically adsorbed surface water were therefore excluded. As shown in the survey spectra (Figure S1), the Si 2p signal is only marginally detectable, indicating that the O 1s response is not dominated by the underlying SiO_2 substrate. Moreover, since all samples were fabricated on the same 100 nm-thick SiO_2 and measured under identical XPS conditions, any residual substrate-related contributions remain substantially constant over t_A and are inadequate to explain the observed O_{NL} changes. Additionally, it is noted that since XPS is inherently surface-sensitive with an information depth of only a few nanometers, ultrathin films can exhibit fine substrate signals while primarily reflecting the chemical state near the surface of In_2O_3. Quantitatively, the O_L fraction increases from 42.40% (t_A = 5 min) to 55.40% (t_A = 30 min), followed by a slight decrease to 52.00% at 180 min.

In contrast, the O_{NL} fraction decreases from 57.60% at 5 min to 44.60% at 30 min and then increases again to 48.00% at 180 min. The initial increase in O_L up to 30 min is consistent with continued network condensation inferred from the XRD results. A higher O_L fraction indicates that a larger proportion of oxygen atoms occupy bonding environments closer to ideal lattice coordination, which corresponds to a reduction in M-OH or incomplete and loosely bonded M-O-M structures during the early annealing stage.

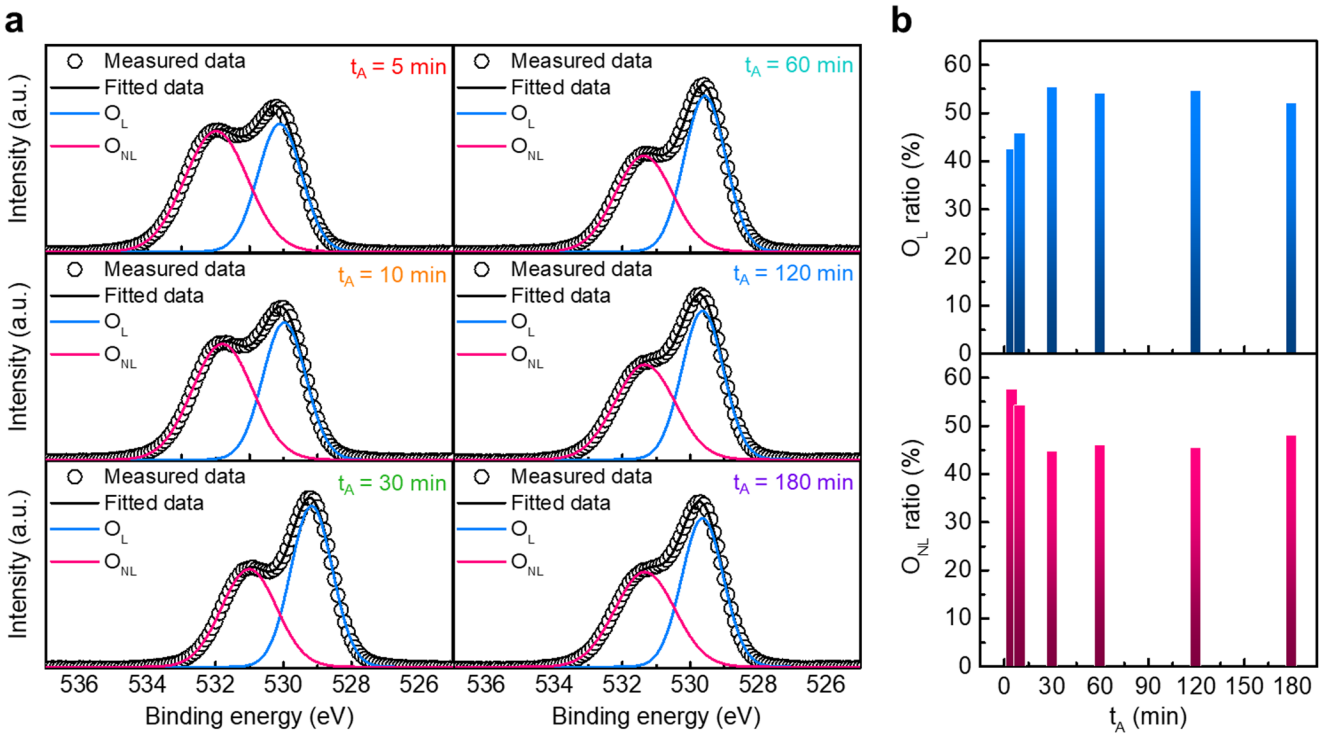


Fig. 3 (**a**) XPS O 1s spectra of UTB In_2O_3 annealed at various t_A, deconvoluted into O_L and O_{NL} components. (**b**) Extracted O_L and O_{NL} fractions as a function of t_A

The decrease in O_L and the rebound in O_{NL} at longer t_A values (60–180 min) represent key observations that should be carefully considered. As mentioned above, it is inappropriate to equate O_{NL} with a single defect type such as bulk V_O without corroborating evidence, because multiple mechanisms (including oxygen-deficient local environments, hydroxyl-related bonding, and other defective oxygen configurations) can contribute to similar binding energy ranges [27–29]. Given the ongoing debate regarding assignment of the O 1s peak at ~531 eV solely to bulk V_O, the O_{NL} rebound with increasing t_A is conservatively interpreted as an increase in O_{NL} near surface or defect-rich regions rather than as a direct signature of bulk V_O formation. Even though prolonged annealing promotes densification and structural relaxation, it may also increase the density of sites of surface sites that facilitate adsorption of H or H_2O, such as under-coordinated In atoms or oxygen-deficient sites [30–32]. XRD primarily probes long-range lattice periodicity, whereas XPS is inherently surface-sensitive and reflects local chemical bonding within the near-surface region. Therefore, although the two techniques provide complementary insights, direct one-to-one correspondence between XRD-derived trends and O 1s spectral components should not be assumed.

To examine the impact of t_A on the electrical characteristics of crystalline UTB In_2O_3/Y_2O_3 TFTs, we measured the transfer characteristics of TFTs with various t_A of In_2O_3 (Fig. 4a). Independent of t_A, all TFTs exhibit clear switching behavior, extremely low gate leakage currents, and off-state drain currents in the range of $10^{-12} - 10^{-10}$ A. These results indicate that crystalline UTB In_2O_3/Y_2O_3 TFTs maintain stable basic operation across the investigated t_A range.

However, the electrical parameters exhibit a distinct dependence on t_A. V_{TH}, SS, and μ_{SAT}, which are key electrical parameters extracted from the transfer characteristics, are shown in Fig. 4b-d, respectively. As t_A increases, V_{TH} shifts monotonically in the negative direction, decreasing from 2.92 V (5 min) to 1.18 V (180 min). A negative V_{TH} shift in In_2O_3 TFTs is typically associated with an increase in the effective electron density in the channel or an enhancement of donor-like characteristics. This behavior can arise from structural relaxation of the In_2O_3 network and from the O_{NL} component related to V_O or -OH species. SS remains in the range of 0.43–0.51 V dec^{-1} up to 60 min, after which it increases to 0.58 V dec^{-1} at 120 min and 0.72 V dec^{-1} at 180 min. The increase in SS is mainly attributed to an increase in the subgap states at the insulator-semiconductor interface or within the semiconductor bulk, as described by the following relation [33]:

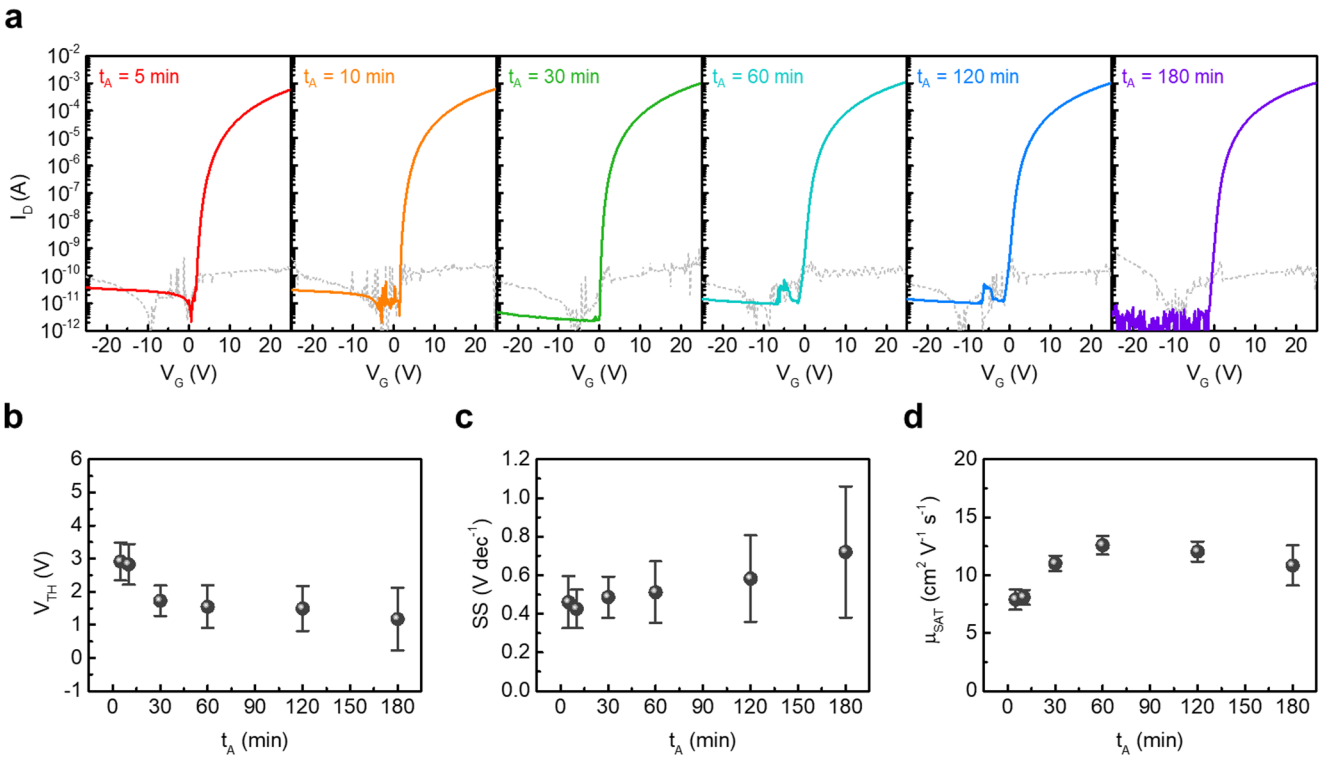


Fig. 4 (**a**) Representative transfer characteristics of UTB In_2O_3 TFTs annealed at various t_A. Statistical comparison of (**b**) V_{TH}, (**c**) SS, and (**d**) μ_{SAT} as a function of t_A

$$SS \approx \frac{kT}{q}\ln 10\left(1+\frac{q\left(t_{ch}N_T+D_{it}\right)}{C_{ox}}\right) \approx \frac{kT}{q}\ln 10\left(1+\frac{qtchN_{T,max}}{C_{ox}}\right) \quad (4)$$

Here, q, k, T, t_{ch}, C_{ox}, N_T, and D_{it} denote the elementary charge, Boltzmann constant, absolute temperature, channel thickness, gate capacitance per unit area, bulk trap density, and interface trap density, respectively.

μ_{SAT} increases from 7.90 $cm^2\ V^{-1}\ s^{-1}$ (5 min) to a maximum of 12.58 $cm^2\ V^{-1}\ s^{-1}$ (60 min) and then decreases to 10.84 $cm^2\ V^{-1}\ s^{-1}$ (180 min) as t_A increases. The interval from 5 to 60 min can therefore be interpreted as a regime in which structural relaxation and crystallite growth stabilize the channel network and primarily enhance electrical performance, consistent with the XRD results. The increase in grain size implies a reduction in grain boundary density and a weakened influence of grain-boundary barriers, which is consistent with the monotonic increase in μ_{SAT} up to 60 min. The relatively low O_{NL} fraction observed between approximately 30 and 60 min further supports this interpretation.

In the 60–180 min range, SS and μ_{SAT} progressively deteriorate, despite the continued decrease in d-spacing and increase in crystallite size observed by XRD. Consistent with the SS expression above, the most conservative interpretation is that the effective subgap density of states increases at longer t_A. The increase in O_{NL} observed by XPS provides a plausible chemical signature of this change. O_{NL} reflects a higher fraction of oxygen-related non-ideal bonding environments, which can introduce additional subgap tails or Gaussian states. Because XPS alone cannot identify specific defect types or their precise energy distributions, it is not appropriate to assign a particular defect level as the direct origin of the SS and μ_{SAT} degradation. Nonetheless, given that excessive t_A coincides with O_{NL} regrowth, it is reasonable to assume contributions from V_O - or -OH-related configurations promoted at longer t_A.

Bias reliability tests were performed to clarify how t_A-modulated structural and chemical properties of UTB In_2O_3 influence device stability. Figure 5a shows the evolution of the transfer curves during NBS (V_{STR} = -40 V). Figure 5b-d summarize ΔV_{TH}, ΔSS, and $\Delta\mu_{SAT}$ under NBS for all t_A conditions. All electrical parameters exhibit negligible changes under NBS, and no clear dependence on t_A is observed. This robust NBS behavior is consistent with the characteristics of n-type oxide semiconductors, which possess extremely low hole concentrations due to their electronic structure. As a result, the channel remains depleted, and hole trapping is not promoted under negative gate bias [34]. Furthermore, the Y_2O_3 capping layer suppresses the adsorption and desorption of atmospheric species that can induce negative bias instability. Consequently, crystalline UTB In_2O_3/Y_2O_3 TFTs exhibit excellent NBS stability across the investigated t_A range.

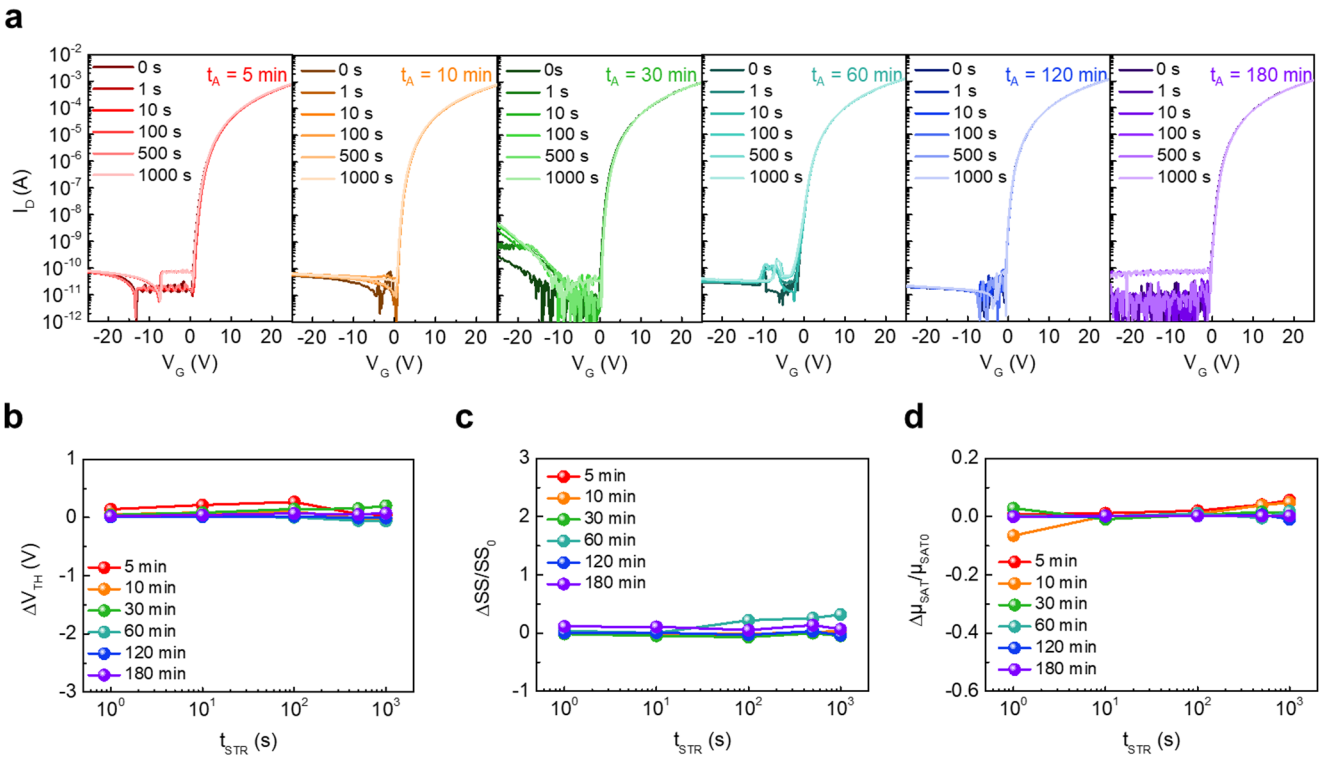


Fig. 5 (**a**) Evolution of transfer curves under NBS for UTB In_2O_3 TFTs annealed at various t_A. (**b**) ΔV_{TH}, (**c**) $\Delta SS/SS_0$, and (**d**) $\Delta\mu_{SAT}/\mu_{SAT0}$ as a function of t_A

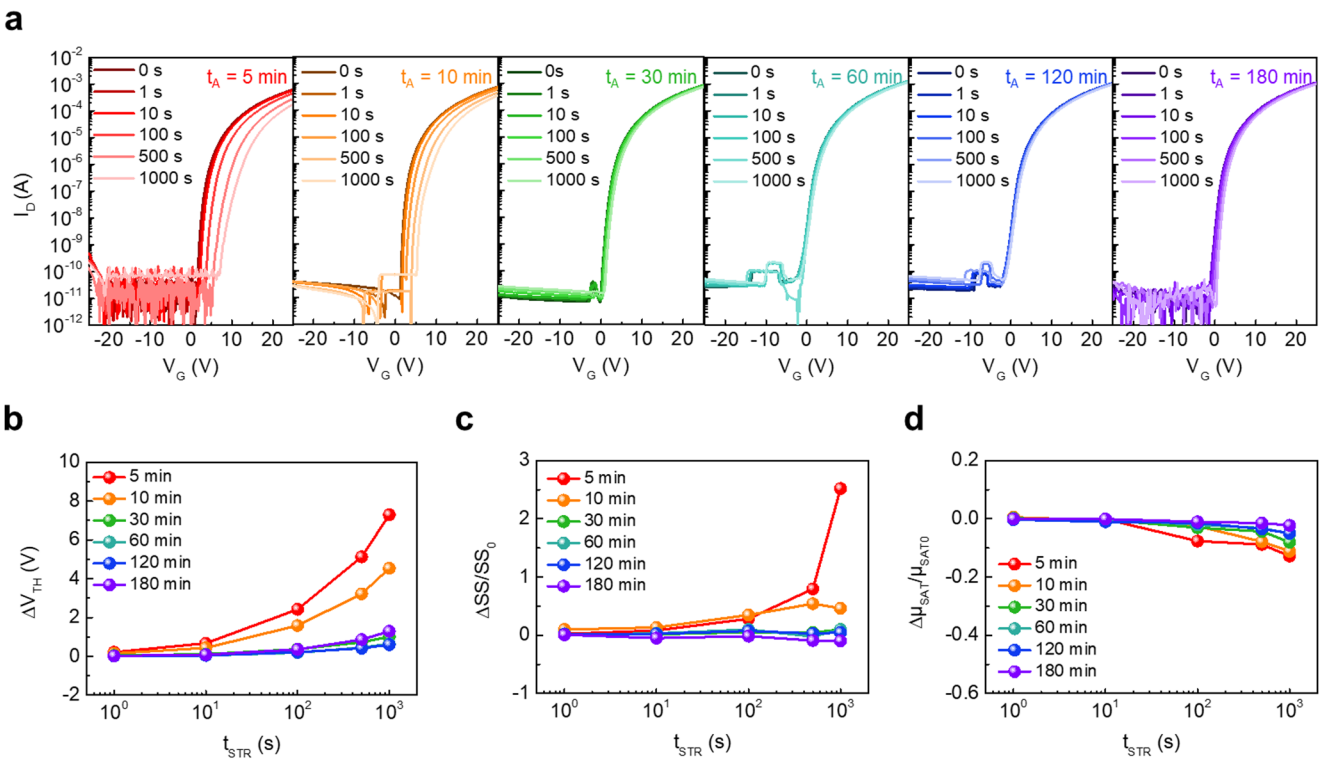


Fig. 6 (**a**) Evolution of transfer curves under PBS for UTB In_2O_3 TFTs annealed at various t_A. (**b**) ΔV_{TH}, (**c**) $\Delta SS/SS_0$, and (**d**) $\Delta\mu_{SAT}/\mu_{SAT0}$ plotted as a function of t_A

In contrast, as shown in Fig. 6a, the evolution of transfer characteristics under PBS (V_{STR} = 40 V) reveals a pronounced dependence on t_A. PBS-induced instability is commonly explained by charge trapping and defect creation mechanisms [35]. Charge trapping mainly induces ΔV_{TH}, whereas field-induced defect generation can simultaneously affect ΔV_{TH}, ΔSS, and $\Delta\mu_{SAT}$.

In this study, ΔV_{TH} under PBS decreases rapidly as t_A increases from short times to 60 min and then approaches saturation at longer t_A. Similarly, ΔSS and $\Delta\mu_{SAT}$ show a gradual improvement from the shortest t_A of 5 min up to 180 min. This monotonic improvement in ΔSS and $\Delta\mu_{SAT}$ is consistent with the structural trends observed by XRD trends. The continuous decrease in d-spacing and increase in crystallite size with t_A suggest ongoing structural relaxation, increased film density, and improved crystalline coherence. A denser and more relaxed In_2O_3 network is expected to contain fewer weakly bonded configurations that can reorganize under high electric fields. Consequently, this prevents defect generation by PBS, which is reflected in the gradual improvement in ΔSS and $\Delta\mu_{SAT}$ with increasing t_A. In addition, O_{NL} directly reflects the subgap state density as a chemical characteristic of the non-ideal oxygen binding environment. Therefore, O_{NL} shows a strong correlation with SS, as confirmed in Fig. 4, which implies its potential influence on PBS instability. However, PBS instability follows the structural relaxation trend and does not show a clear correlation with O_{NL} regrowth in this study. This observation suggests that structural relaxation and crystallite growth are more dominant factors in the PBS instability of the crystalline UTB TFTs examined here.

Although elevated-temperature PBS and/or NBS can provide additional insight into thermally activated kinetics, the present PBS evaluation was performed under a stringent vertical electric field of 4 MV cm^{-1}, which sufficiently accelerates field-assisted degradation even at room temperature [36]. Therefore, the stability trend with t_A is expected to be qualitatively maintained although the characteristic time constant is mainly shortened as the temperature increases. However, since stress temperature can introduce additional degradation mechanisms or alter the influence of degradation factors, more in-depth study is required [37].

Overall, an in-depth understanding of the t_A-dependent structural, chemical, and electrical characteristics of UTB In_2O_3 suggests that UTB In_2O_3 exhibits an annealing-time-dependent process window for realizing desirable TFT performance. The intermediate t_A region (approximately 30–60 min) exhibits low SS while maximizing μ_{SAT} and maintaining excellent NBS and PBS stability. Excessively short t_A may improve throughput but can result in degraded electrical characteristics and reliability. Conversely, excessively long t_A can promote increased O_{NL} due

to over-annealing, which leads to degradation of u_{SAT} and SS. Furthermore, structural relaxation or densification and oxygen-related defect evolution can occur simultaneously in the UTB oxide channel but affect different device properties, which results in intrinsic trade-offs between performance and stability.

4 Conclusions

This work demonstrates that t_A is a decisive process parameter that simultaneously governs densification, oxygen-bonding evolution, and bias stability in UTB In_2O_3 TFTs, which leads to a clear performance–stability trade-off rather than monotonic optimization. As t_A increases, XRD consistently indicates progressive structural relaxation, with d-spacing contraction and crystallite size growth, which implies improved lattice ordering and densification. However, O 1s XPS reveals a reversal in the oxygen-bonding trend at long t_A, which is conservatively interpreted as an increase in O_{NL} environments in near-surface or defect-rich regions rather than as a single-defect signature. This O_{NL} rebound provides a consistent chemical context for the electrical degradation observed at long t_A, where SS increases from 0.45 to 0.51 V dec^{-1} ($t_A \leq 60$ min) to 0.72 V dec^{-1} ($t_A = 180$ min) and μ_{SAT} decreases from a peak of 12.58 $cm^2 V^{-1} s^{-1}$ ($t_A = 60$ min) to 10.84 $cm^2 V^{-1} s^{-1}$ ($t_A = 180$ min). Reliability results further show that negative bias stress is intrinsically stable in these n-type UTB channels and is reinforced by the Y_2O_3 capping layer, whereas positive bias stress instability is progressively mitigated with increasing t_A, consistent with defect suppression driven by structural relaxation and reduced field-activated trapping or defect creation. Collectively, these results identify annealing-time-modulated structural relaxation and O_{NL} evolution as the primary origin of the performance–stability trade-off in UTB TFTs. An optimal processing window is therefore established to maximize electrical performance while maintaining robust PBS stability.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

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